



2025

M23(B)

BOOKLET NO.

217021

Electrical Engineering
Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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SECTION – A

Q1. Answer the following. (10×5=50)

Q1. (a) State and explain the types of control system with suitable examples. 10

Q1. (b) Determine the contents of accumulator (A), Program Status Word (PSW) and Program Counter (PC) of microprocessor 8085 after the execution of the instruction ADDM which is stored at memory location 7050h. Given that the contents of various components of 8085 are, (A) = 2Fh, (H) = 25h, (L) = 00h, (PSW) = 2Ah and that of the memory location 2500h is A8h. 10

Q1. (c) Explain “Two Wattmeter” method for measurement of power in 3-phase circuit with Y-connected balanced load and show that the total active power is the sum of two wattmeter readings. 10

Q1. (d) Obtain the overall transfer function of the system shown in Fig. 1(d). 10

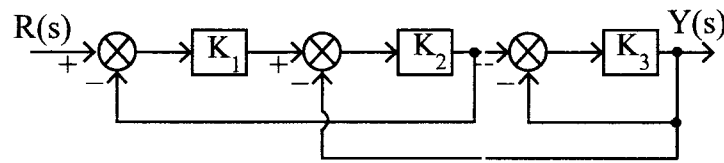


Fig. 1(d)

Q1. (e) What are the various addressing modes of 8085 ? Explain them with atleast two examples from the instruction set. 10

Q2. (a) For the electrical circuit shown in Fig. 2(a).

i) Identify the set of state variables.

ii) Draw a signal flow graph of the circuit in terms of the state variables as identified above.

iii) From the signal flow graph, write the state variable equations of the circuit.



- iv) From the signal flow graph, determine the transfer function in S-domain. 20

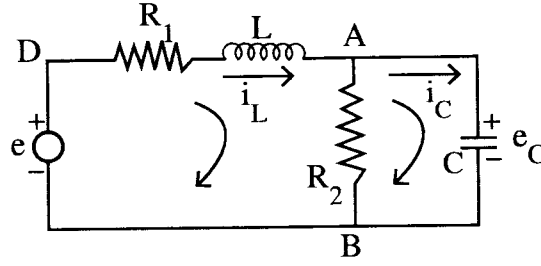


Fig. 2(a)

- Q2. (b) Explain the timing diagram for Op-code fetch operation for the microprocessor 8085 with an example to fetch the Op-code "OF" stored at memory location 6050h. 20

- Q2. (c) The open loop transfer function of unity feedback system is given as,
 $G(s) = \frac{K}{s(s+1)}$. Determine the percentage of "K" to be reduced so that the peak overshoot is reduced from 30% to 20%. 10

- Q3. (a) i) Using R-H criteria, find the number of poles in the left half-plane, the right half-plane and on the $j\omega$ -axis for the system shown in Fig. 3(a) (i). 10

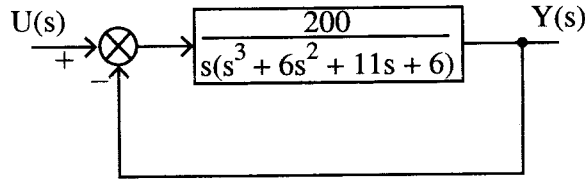


Fig. 3(a) (i)

- Q3. (a) ii) Determine the steady state error for the unit step input applied to the system shown in Fig. 3(a) (ii). 10

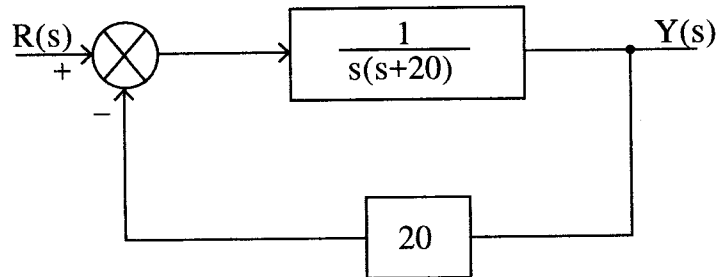


Fig. 3(a) (ii)



Q3. (b) Explain in brief the following transducers with their functions:

- i) Thermistor
- ii) Piezoelectric crystal
- iii) Strain Gauge
- iv) LVDT

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Q3. (c) For a system, transfer function is given by $\frac{K}{1 + \frac{s}{a}}$ and the asymptotic Bode Plot is as shown in Fig. 3(c). Determine the error in phase angle and gain (dB) for

the exact and asymptotic plots.

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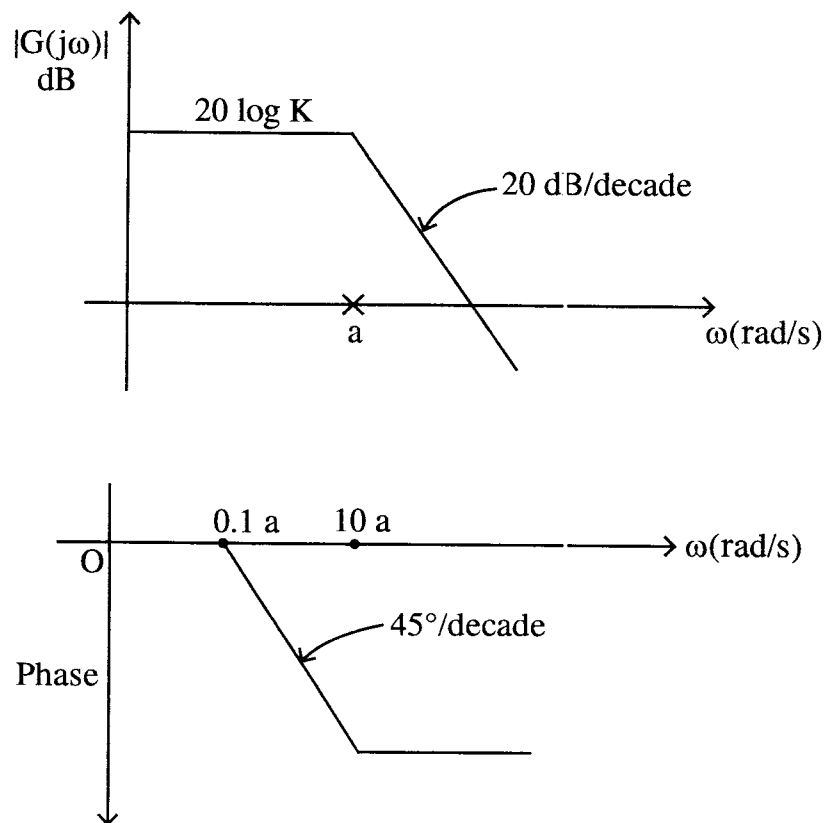


Fig. 3(c)



- Q4. (a)** Explain Proportional (P) control and Integral (I) control. Use these controllers independently for the system given by, $G(s) = \frac{2}{(s+1)(s+2)}$ to compare the proportional error constant (K_p) and steady state errors. Consider the value of static gain block (K) as 50. **20**
- Q4. (b)** Explain with a block diagram the Programmable Peripheral Interface (8255) device. Also describe the various operating modes of 8255. State the mode of operation and form the control word to make the pin number 3 of port C as high ($PC_3 = 1$). Consider don't care pins as "1". **20**
- Q4. (c)** Draw and explain Maxwell's Inductance – Capacitance bridge for measurement of unknown inductance and obtain its equation at balanced condition of bridge. **10**
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SECTION – B

- Q5.** Answer the following. **(10×5=50)**
- Q5. (a) i)** The p.u. impedance of a transmission line is 0.2. What will be new p.u. value of the impedance, if base kV and base MVA are doubled ? **5**
- Q5. (a) ii)** Determine the capacitance and charging current per km when a transmission line is operating at 50 Hz and 132 kV. Given that, radius of conductor is 0.4 cm and the mutual GMD of conductor (D_m) is 2.015 meters.
(Assume $\epsilon_0 = 8.854 \times 10^{-12}$ F/m) **5**
- Q5. (b)** A 3-phase Star-Delta power transformer is connected in a distribution system. Suggest and describe the suitable protection scheme against the internal faults, with neat diagram showing the connections of CTs in primary and secondary sides of transformer. **10**
- Q5. (c)** Explain pulse code modulation. What are various steps involved in PCM ? **10**
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- Q5. (d) For the power system shown in Fig. 5(d), the p.u. reactances of various components are marked. Assuming the pre-fault voltage as 1.0 p.u., calculate the fault current I_f and the voltages at the Bus 1 and Bus 2 for a solid 3-phase fault on Bus 3.

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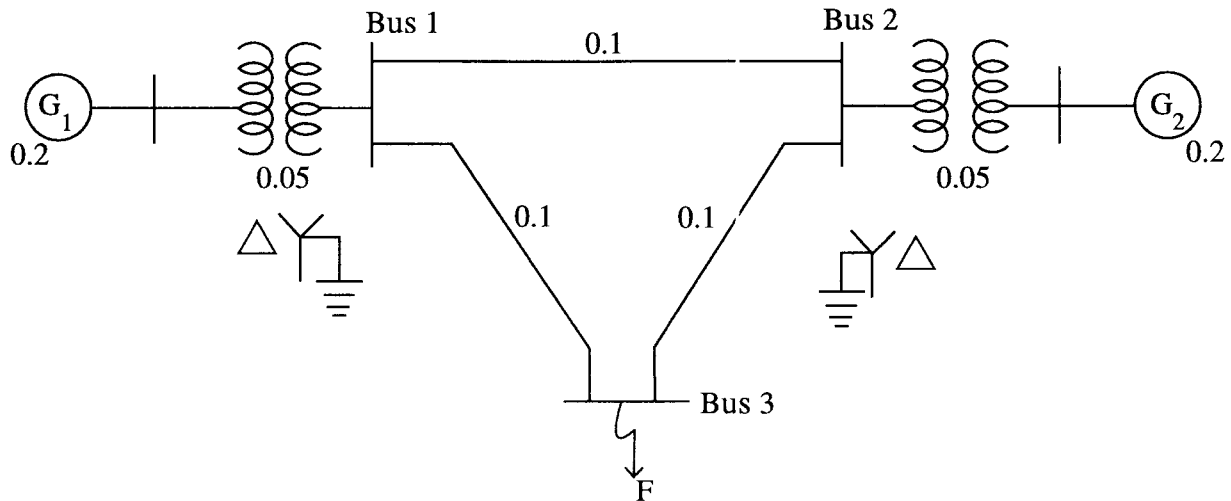


Fig. 5(d)

- Q5. (e) Explain the phenomenon of current chopping in circuit breakers. Also state the remedy to overcome the adverse effect due to current chopping.

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- Q6. (a) A 220 kV, 3-phase transmission line is 40 km long. The resistance per phase is 1.3263 mH per km. Assume shunt capacitance is negligible and the line model to be “short line” model. Determine the voltage and power at sending end. Also determine the voltage regulation and efficiency. Obtain these quantities independently when the line is supplying 3-phase load of

i) 381 MVA at 0.8 p.f. lagging at 220 kV.

ii) 381 MVA at 0.8 p.f. leading at 220 kV.

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- Q6. (b) i) For an IDMT relay, a typical characteristic with TMS = 1.0, is given by the table 6(b) (i). The relay is having a rating of 5 A and is connected to a circuit through CT of ratio 500/5. If the fault current is 5000 A, determine the time of operation of the relay having a setting of 125% TMS = 0.6.

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Current as multiple of PSM	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0	20.0
Time (Seconds)	100	10	6.2	5.4	4.7	4.1	3.6	3.2	2.9	2.8	2.2

Table 6(b) (i)



- Q6. (b) ii) In a short circuit test on 3-pole, 132 kV C.B., the following observations are made – (i) p.f. of the fault is 0.6, (ii) the breaking current is symmetrical, (iii) the recovery voltage is same as line voltage, (iv) the frequency of oscillations is 16 kHz. Assuming that the neutral is grounded and fault does not involve ground, determine the average rate of rise of restriking voltage. 10
- Q6. (c) What are the various digital modulation techniques ? Explain them in brief. 10

- Q7. (a) A 50 Hz synchronous generator having inertia constant $H = 5$ MJ/MVA and a direct axis transient reactance $X'_d = 0.3$ p.u. is connected to an infinite bus through two parallel lines as shown in Fig. 7(a). Reactances are marked at the respective components, on a common system base. The generator is delivering real power $P_e = 0.8$ p.u. and $Q = 0.074$ p.u. to the infinite bus at a voltage of $V = 1$ p.u. Determine :

- i) Critical clearing angle and critical fault clearing time when a temporary 3-phase fault occurs at the sending end point and is cleared to remain the lines intact.
- ii) Critical clearing angle when a 3- phase fault occurs at middle of one of the lines and is cleared with the isolation of the faulty line. 20

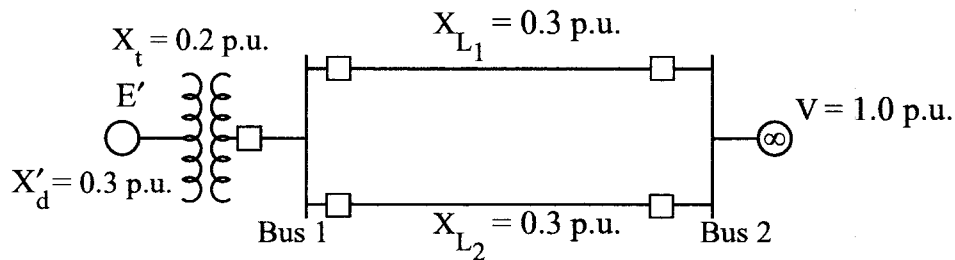


Fig. 7(a)



Q7. (b) State and prove the sampling theorem.

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Q7. (c) The fuel inputs per hour of plants 1 and 2 are given as

$$F_1 = 0.2 P_1^2 + 40 P_1 + 120 \text{ Rs/Hr}$$

$$\text{and } F_2 = 0.25 P_2^2 + 30 P_2 + 150 \text{ Rs/Hr}$$

Determine the economic operating schedule and the corresponding cost of generation if the maximum and minimum loading on each unit is 120 MW and 40 MW. The total demand is 160 MW and transmission losses are neglected.

If the load is equally shared by both the units, determine also the saving obtained by loading the units as per equal incremental production cost.

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Q8. (a) Draw HVDC convertor station layout. What are the various components of HVDC convertor station ? Explain them in brief.

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Q8. (b) Explain with block diagram, the microprocessor based distance protection to protect a transmission line. Also draw the flow chart algorithm to carry out the required protection.

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Q8. (c) The generator shown in the following Fig. 8(c), delivers an active power 1.0 p.u. to infinite bus. Considering the terminal voltage of generator as 1.0 p.u., determine the maximum power which can be transferred to infinite bus when,

i) System has no fault.

ii) One of the two parallel lines is open.

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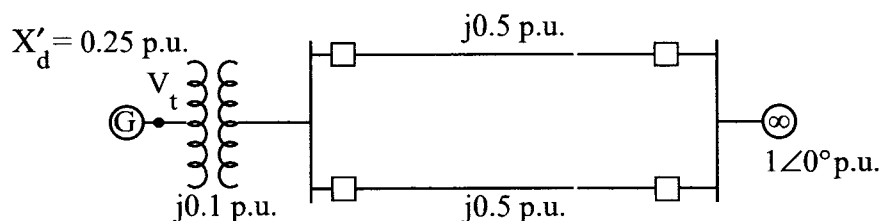


Fig. 8(c)